Shruti Sharma

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EDUCATION

Indian Institute of Technology (IIT) Mandi *Himachal Pradesh, India*
B.Tech in Data Science Engineering 2024 – Present

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Computer Organisation, Software Design

TECHNICAL SKILLS

Languages: Python, C++

Web & Systems: REST APIs, Git/GitHub

AI / ML & NLP: Machine Learning, Reinforcement Learning, NLP, TF-IDF, Statistical Modeling,

Frameworks & Libraries: scikit-learn, Streamlit, Plotly

Concepts: Data Structures & Algorithms, Object-Oriented Programming, Software Design, Problem Solving

PROJECTS

NewsLens – Hyper-Personalization Engine *2nd Place, HACK60 (IIT Mandi × HCLTech)*
Python, Reinforcement Learning, TF-IDF, LLMs, REST APIs

- Engineered a hybrid recommendation system combining Thompson Sampling reinforcement learning, TF-IDF, and collaborative filtering to rank and personalize content for users at scale, following modular, testable software design practices.
- Trained and validated the pipeline on 156K user interactions from the industry-standard MIND dataset, quantifiably improving content relevance and ranking accuracy.
- Built a real-time LLM re-ranking module (Groq / Llama 3 via REST API) using prompt engineering techniques to dynamically re-score and re-order recommendations based on live signal inputs.
- Optimized the end-to-end inference pipeline to deliver sub-500ms latency — 4x faster than the target benchmark — reflecting a focus on efficient, scalable, production-grade system design.

AI-Driven Mock Interview System GitHub
Python, Streamlit, NLP, Full-Stack Deployment

- Built and deployed a full-stack web application with an automated question-generation engine that parses candidate resumes and dynamically produces role-relevant interview questions using NLP.
- Implemented TF-IDF and cosine-similarity-based scoring to systematically evaluate response quality against expected answers, removing manual grading bottlenecks.
- Designed a modular four-stage architecture (resume parsing → question generation → answer evaluation → automated report generation), applying clean software design principles for end-to-end reliability.
- Shipped the application to production for public, real-time access, supporting concurrent users in live interview practice sessions.

Resume ATS Analyzer GitHub
Python, Streamlit, scikit-learn, Web Deployment

- Developed and deployed an NLP-based scoring engine as a live web application to benchmark resumes against job descriptions and automatically flag missing skills, improving decision-making speed for end users.
- Built an interactive full-stack analytics dashboard to visualize match scores, with exportable reports summarizing quantitative fit.
- Iterated on the product end-to-end — from data processing logic to UI — enabling recruiters and candidates to interactively compare resumes against job requirements.

ACHIEVEMENTS

- 2nd Place, HACK60 — Hackathon organized by IIT Mandi in collaboration with HCLTech.